MCQ Questions and Answers on Design and Model Database

Part 1: Database Fundamentals

1. What is a database?
A. A collection of programs
B. An organized collection of data
C. An operating system
D. A computer network
Answer: B
An organized collection of data.

2. What is DBMS?
A. Data Backup Management System
B. Database Management System
C. Digital Base Management Software
D. Database Monitoring Service
Answer: B

3. Which of the following is a popular relational database management system?
A. Photoshop
B. MySQL
C. MS Paint
D. PowerPoint
Answer: B

4. What is the primary purpose of database design?
A. To increase file size
B. To organize data efficiently
C. To slow down data access
D. To delete unnecessary software
Answer: B

5. What is a database model?
A. A physical computer
B. A structure that defines how data is organized
C. A programming language
D. An operating system
Answer: B

6. Which database model stores data in tables?
A. Hierarchical model
B. Network model
C. Relational model
D. Object-oriented model
Answer: C

7. What is a table in a relational database?
A. Collection of rows and columns
B. Collection of programs
C. Collection of networks
D. Collection of files only
Answer: A

8. What is a row in a database table called?
A. Field
B. Record
C. Attribute
D. Column
Answer: B

9. What is a column in a database table called?

A. Record
B. Tuple
C. Attribute
D. Database
Answer: C
10. Which language is commonly used to manage relational databases?
A. HTML
B. CSS
C. SQL
D. XML
Answer: C

Part 2: ER Diagram and Database Modeling

11. What does ERD stand for?
A. Entity Relationship Diagram
B. Electronic Record Design
C. Entity Record Database
D. External Relationship Data
Answer: A
12. What is an entity in database design?
A. A database query
B. A real-world object or concept
C. A programming command
D. A database password
Answer: B
13. Which symbol represents an entity in an ER diagram?
A. Circle
B. Rectangle
C. Diamond
D. Triangle
Answer: B
14. Which symbol represents an attribute in a traditional ER diagram?
A. Oval
B. Rectangle
C. Diamond
D. Square
Answer: A
15. Which symbol represents a relationship in a traditional ER diagram?
A. Circle
B. Rectangle
C. Diamond
D. Line
Answer: C
16. What is a relationship in database modeling?
A. Connection between entities
B. Database password
C. Data type
D. SQL command
Answer: A
17. What does a one-to-one relationship mean?
A. One record relates to many records
B. Many records relate to many records

C. One record relates to one record
D. No records are related
Answer: C
18. What does a one-to-many relationship mean?
A. One record relates to multiple records
B. One record relates to one record
C. Multiple records relate to one record only
D. No relationship exists
Answer: A
19. What does a many-to-many relationship mean?
A. One entity has no attributes
B. Multiple records relate to multiple records
C. One record relates to one record
D. Data cannot be stored
Answer: B
20. Which tool is commonly used to create ER diagrams?
A. MySQL Workbench
B. VLC Player
C. Notepad
D. Windows Media Player
Answer: A
Part 3: Keys and Constraints
21. What is a primary key?
A. A key that allows duplicate values
B. A unique identifier for each record
C. A temporary password
D. A database name
Answer: B
22. Can a primary key contain NULL values?
A. Yes
B. No
C. Sometimes
D. Only in MySQL
Answer: B
23. What is a foreign key?
A. A key used to connect related tables
B. A key used to delete databases
C. A key used to create users
D. A key used to encrypt data
Answer: A
24. What is a candidate key?
A. A key that uniquely identifies records
B. A key that contains duplicate values
C. A key that allows NULL values only
D. A key used for sorting
Answer: A
25. What is a composite key?
A. A key containing multiple columns
B. A key containing one column only
C. A key used for backup
D. A key used for encryption
Answer: A

26. Which constraint prevents duplicate values in a column?
A. CHECK
B. UNIQUE
C. DEFAULT
D. NOT NULL
Answer: B
27. Which constraint prevents NULL values?
A. UNIQUE
B. CHECK
C. NOT NULL
D. DEFAULT
Answer: C
28. Which constraint validates data according to a condition?
A. CHECK
B. DEFAULT
C. PRIMARY KEY
D. INDEX
Answer: A
29. What is the purpose of referential integrity?
A. To maintain consistency between related tables
B. To increase duplicate records
C. To delete all tables
D. To change database names
Answer: A
30. Which key can contain duplicate values?
A. Primary key
B. Candidate key
C. Foreign key
D. Unique key
Answer: C

Part 4: Database Normalization

31. What is database normalization?
A. Organizing data to reduce redundancy
B. Increasing duplicate data
C. Deleting all records
D. Encrypting database tables
Answer: A
32. What is the main purpose of normalization?
A. Reduce data redundancy
B. Increase data duplication
C. Increase storage usage
D. Slow down queries
Answer: A
33. What does 1NF stand for?
A. First Normal Form
B. First Network Format
C. First Number Function
D. First Name Field
Answer: A
34. Which normal form requires atomic values?
A. 1NF
B. 2NF

C. 3NF
D. BCNF
Answer: A
35. What is the main requirement of 2NF?
A. Must be in 1NF and have no partial dependency of non-key attributes on candidate keys
B. Must contain duplicate values
C. Must have no primary key
D. Must contain NULL values
Answer: A
36. What is the main purpose of 3NF?
A. Remove transitive dependencies
B. Increase duplicate records
C. Remove all primary keys
D. Remove all foreign keys
Answer: A
37. Which normal form is stricter than 3NF?
A. 1NF
B. 2NF
C. BCNF
D. None
Answer: C
38. What is data redundancy?
A. Repetition of the same data unnecessarily
B. Data encryption
C. Data backup
D. Data deletion
Answer: A
39. Which problem can occur due to data redundancy?
A. Update anomaly
B. Faster processing always
C. Better security
D. Automatic backup
Answer: A
40. What is denormalization?
A. Combining data to improve read performance
B. Removing all database tables
C. Deleting primary keys
D. Encrypting records
Answer: A

Part 5: Database Design and SQL

41. Which SQL command is used to create a database?
A. INSERT
B. CREATE DATABASE
C. UPDATE
D. SELECT
Answer: B
42. Which SQL command is used to create a table?
A. CREATE TABLE
B. DELETE TABLE
C. SELECT TABLE
D. UPDATE TABLE
Answer: A

43. Which SQL data type is suitable for storing text of variable length?
A. INT
B. DATE
C. VARCHAR
D. FLOAT
Answer: C
44. Which data type is suitable for storing date values?
A. INT
B. DATE
C. VARCHAR
D. BOOLEAN
Answer: B
45. Which SQL command modifies the structure of an existing table?
A. UPDATE
B. ALTER TABLE
C. INSERT
D. SELECT
Answer: B
46. Which command is used to create an index?
A. CREATE INDEX
B. INSERT INDEX
C. UPDATE INDEX
D. SELECT INDEX
Answer: A
47. What is the purpose of an index in a database?
A. To improve data retrieval speed
B. To delete records
C. To increase duplicate data
D. To remove constraints
Answer: A
48. Which database design level describes how data is physically stored?
A. Conceptual level
B. Logical level
C. Physical level
D. External level
Answer: C
49. Which database design level describes entities, attributes, and relationships?
A. Conceptual level
B. Physical level
C. Hardware level
D. Network level
Answer: A
50. Which of the following is an important step in database design?
A. Requirement analysis
B. Deleting all data
C. Installing games
D. Formatting the computer
Answer: A

MCQ Questions and Answers on Set Up Database Environment

Part 1: Database Environment Fundamentals

1. What is a database environment?
A. A programming language
B. A combination of hardware, software, data, and users
C. An operating system only
D. A network cable
Answer: B

2. What does DBMS stand for?
A. Data Backup Management System
B. Database Management System
C. Digital Base Management Software
D. Database Monitoring Service
Answer: B

3. Which of the following is a DBMS software?
A. MySQL
B. Photoshop
C. VLC Player
D. MS Paint
Answer: A

4. Which of the following is an open-source DBMS?
A. Microsoft SQL Server
B. Oracle Database
C. MySQL
D. Microsoft Access
Answer: C

5. What is the primary purpose of setting up a database environment?
A. To play games
B. To create, manage, and maintain databases
C. To edit images
D. To browse websites
Answer: B

6. Which operating system can support MySQL Server?
A. Windows
B. Linux
C. macOS
D. All of the above
Answer: D

7. Which component is required to run a database server?
A. Hardware
B. Database software
C. Operating system
D. All of the above
Answer: D

8. Which of the following is a database server?
A. MySQL Server
B. Google Chrome
C. MS Word
D. Paint
Answer: A

9. What is the default port number of MySQL Server?

A. 80
B. 21
C. 3306
D. 443
Answer: C
10. Which tool is commonly used to manage MySQL databases graphically?
A. MySQL Workbench
B. Notepad
C. Calculator
D. Windows Media Player
Answer: A
Part 2: Database Installation and Configuration
11. What is the first step in setting up a database environment?
A. Delete database
B. Check system requirements
C. Create a table
D. Insert records
Answer: B
12. Which software package can install MySQL Server on Windows?
A. MySQL Installer
B. Adobe Reader
C. WinRAR
D. VLC Player
Answer: A
13. What is the purpose of MySQL Installer?
A. To install and configure MySQL products
B. To design images
C. To create presentations
D. To browse the Internet
Answer: A
14. What is the purpose of configuring a root password?
A. To improve screen resolution
B. To secure administrative database access
C. To increase RAM
D. To create a network cable
Answer: B
15. Which command checks the MySQL version?
A. SHOW VERSION;
B. SELECT VERSION();
C. CHECK VERSION;
D. GET VERSION;
Answer: B
16. Which command is used to start the MySQL service in Windows Command Prompt?
A. NET START MySQL80
B. NET DELETE MySQL80
C. NET REMOVE MySQL80
D. NET STOP MySQL80
Answer: A
17. Which command is used to stop the MySQL service in Windows?
A. NET START MySQL80
B. NET STOP MySQL80

C. NET CREATE MySQL80
D. NET RUN MySQL80
Answer: B
18. Which file is commonly used for MySQL server configuration on Windows?
A. my.ini
B. index.html
C. style.css
D. config.php
Answer: A
19. What is the purpose of the MySQL configuration file?
A. To configure server settings
B. To create images
C. To edit documents
D. To play videos
Answer: A
20. Which command connects to MySQL using the command line?
A. mysql -u root -p
B. connect.exe
C. start mysql.exe
D. database -u root
Answer: A

Part 3: Database Creation and Management

21. Which SQL command creates a new database?
A. INSERT DATABASE
B. CREATE DATABASE
C. UPDATE DATABASE
D. DELETE DATABASE
Answer: B
22. Which command displays all databases?
A. SHOW DATABASES;
B. LIST TABLES;
C. DISPLAY DATABASE;
D. SELECT DATABASES;
Answer: A
23. Which command selects a database for use?
A. OPEN database_name;
B. USE database_name;
C. SELECT database_name;
D. START database_name;
Answer: B
24. Which command creates a new table?
A. CREATE TABLE
B. INSERT TABLE
C. ADD DATABASE
D. SELECT TABLE
Answer: A
25. Which command displays all tables in the selected database?
A. SHOW TABLES;
B. SHOW DATABASES;
C. DISPLAY RECORDS;
D. SELECT TABLES;
Answer: A

26. Which SQL command displays the structure of a table?
A. DESCRIBE table_name;
B. DELETE table_name;
C. DROP table_name;
D. INSERT table_name;
Answer: A
27. Which command deletes a database permanently?
A. REMOVE DATABASE
B. DROP DATABASE
C. DELETE DATABASE
D. CLEAR DATABASE
Answer: B
28. Which command is used to insert data into a table?
A. INSERT INTO
B. UPDATE
C. ALTER
D. DROP
Answer: A
29. Which command retrieves data from a table?
A. INSERT
B. SELECT
C. DELETE
D. CREATE
Answer: B
30. Which SQL statement is used to change existing records?
A. UPDATE
B. CREATE
C. DROP
D. SHOW
Answer: A

Part 4: Database Security and User Management

31. Which SQL command creates a database user?
A. CREATE USER
B. ADD DATABASE
C. INSERT USER
D. NEW LOGIN
Answer: A
32. Which SQL command gives privileges to a user?
A. REVOKE
B. GRANT
C. DELETE
D. DROP
Answer: B
33. Which SQL command removes privileges from a user?
A. GRANT
B. CREATE
C. REVOKE
D. INSERT
Answer: C
34. Which account has administrative privileges by default in MySQL?
A. Guest
B. Root

C. Student
D. Public
Answer: B
35. What is the purpose of database authentication?
A. To verify user identity
B. To delete records
C. To increase storage
D. To create tables
Answer: A
36. Which command displays MySQL user accounts?
A. SELECT User, Host FROM mysql.user;
B. SHOW ALL USERS;
C. DISPLAY USERS;
D. LIST ACCOUNT;
Answer: A
37. What is the purpose of a firewall in a database environment?
A. To control network traffic
B. To create tables
C. To format records
D. To increase database size
Answer: A
38. Which is a good database security practice?
A. Sharing root passwords
B. Using strong passwords
C. Giving all users full privileges
D. Disabling authentication
Answer: B

Part 5: Database Connectivity, Backup and Troubleshooting

39. Which protocol is commonly used for database client-server communication over a network?
A. TCP/IP
B. FTP only
C. SMTP only
D. HTTP only
Answer: A
40. Which MySQL utility is used to create a logical backup?
A. mysqldump
B. mysqladmin only
C. ping
D. ipconfig
Answer: A
41. Which command is used to back up a MySQL database from the command line?
A. mysqldump -u root -p database_name > backup.sql
B. DROP DATABASE database_name;
C. DELETE DATABASE database_name;
D. SHOW DATABASES;
Answer: A
42. Which command-line utility is commonly used to restore a MySQL SQL backup?
A. mysql
B. ping
C. ipconfig
D. format
Answer: A

43. What does localhost mean in database connectivity?
A. A remote computer
B. The local computer
C. A cloud server only
D. A network printer
Answer: B
44. Which command checks network connectivity to a server?
A. ping
B. mkdir
C. copy
D. rename
Answer: A
45. What is the purpose of database backup?
A. To recover data after loss
B. To increase duplicate records
C. To delete tables
D. To stop the database server
Answer: A
46. What should you check if MySQL Server fails to start?
A. Service status
B. Configuration file
C. Port conflict and error logs
D. All of the above
Answer: D
47. Which command displays the status of the MySQL server?
A. STATUS;
B. DELETE STATUS;
C. REMOVE STATUS;
D. STOP STATUS;
Answer: A
48. What is the purpose of MySQL Workbench?
A. Database administration and development
B. Video editing
C. Graphic designing
D. Operating system installation
Answer: A
49. Which of the following helps improve database performance?
A. Proper indexing
B. Unnecessary duplicate data
C. Weak passwords
D. Deleting configuration files
Answer: A
50. What is the main responsibility of a database administrator (DBA)?
A. Database installation and maintenance
B. Database security
C. Backup and recovery
D. All of the above
Answer: D

MCQ Questions and Answers on Create and Manage Database Objects

Part 1: Database Objects Fundamentals

1. What is a database object?
A. A hardware device
B. An object used to store or manage database information
C. An operating system
D. A network cable
Answer: B

2. Which of the following is a database object?
A. Table
B. View
C. Index
D. All of the above
Answer: D

3. Which SQL command is used to create a database?
A. INSERT DATABASE
B. CREATE DATABASE
C. UPDATE DATABASE
D. SELECT DATABASE
Answer: B

4. Which SQL command is used to delete a database?
A. DELETE DATABASE
B. REMOVE DATABASE
C. DROP DATABASE
D. CLEAR DATABASE
Answer: C

5. Which SQL command creates a new table?
A. CREATE TABLE
B. ADD TABLE
C. INSERT TABLE
D. UPDATE TABLE
Answer: A

6. Which command displays all tables in the current database?
A. SHOW DATABASES;
B. SHOW TABLES;
C. DISPLAY TABLES;
D. LIST DATABASES;
Answer: B

7. Which command displays the structure of a table?
A. SHOW STRUCTURE
B. DESCRIBE table_name;
C. DISPLAY TABLE
D. SELECT STRUCTURE
Answer: B

8. Which command modifies an existing table structure?
A. UPDATE TABLE
B. ALTER TABLE
C. MODIFY DATABASE
D. CHANGE DATABASE
Answer: B

9. Which command deletes a table permanently?

A. DELETE TABLE
B. REMOVE TABLE
C. DROP TABLE
D. CLEAR TABLE
Answer: C
10. Which command is used to rename a table in MySQL?
A. RENAME TABLE
B. CHANGE TABLE
C. UPDATE TABLE
D. MODIFY TABLE
Answer: A
Part 2: Tables, Columns and Constraints
11. Which data type stores integer values?
A. VARCHAR
B. INT
C. DATE
D. TEXT
Answer: B
12. Which data type stores variable-length character strings?
A. INT
B. DATE
C. VARCHAR
D. FLOAT
Answer: C
13. Which constraint uniquely identifies each record in a table?
A. FOREIGN KEY
B. PRIMARY KEY
C. CHECK
D. DEFAULT
Answer: B
14. Which constraint prevents NULL values in a column?
A. UNIQUE
B. CHECK
C. NOT NULL
D. DEFAULT
Answer: C
15. Which constraint prevents duplicate values in a column?
A. UNIQUE
B. CHECK
C. DEFAULT
D. NOT NULL
Answer: A
16. Which constraint establishes a relationship between two tables?
A. PRIMARY KEY
B. FOREIGN KEY
C. DEFAULT
D. CHECK
Answer: B
17. Which constraint validates data based on a condition?
A. CHECK
B. UNIQUE

C. DEFAULT
D. INDEX
Answer: A
18. Which constraint assigns a default value to a column?
A. PRIMARY KEY
B. FOREIGN KEY
C. DEFAULT
D. UNIQUE
Answer: C
19. Which SQL command adds a new column to an existing table?
A. ALTER TABLE ADD COLUMN
B. INSERT COLUMN
C. CREATE COLUMN
D. UPDATE COLUMN
Answer: A
20. Which SQL command removes a column from an existing table?
A. DELETE COLUMN
B. ALTER TABLE DROP COLUMN
C. REMOVE COLUMN
D. DROP DATABASE
Answer: B
Part 3: Views and Indexes
21. What is a view in a database?
A. A physical table only
B. A virtual table based on a SQL query
C. A database password
D. A backup file
Answer: B
22. Which SQL command creates a view?
A. CREATE VIEW
B. MAKE VIEW
C. INSERT VIEW
D. ADD VIEW
Answer: A
23. Which SQL command deletes a view?
A. DELETE VIEW
B. DROP VIEW
C. REMOVE VIEW
D. CLEAR VIEW
Answer: B
24. What is the main purpose of a database view?
A. To simplify data access
B. To increase duplicate data
C. To delete records
D. To format the hard disk
Answer: A
25. Which SQL statement displays data from a view?
A. INSERT
B. SELECT
C. DROP
D. ALTER
Answer: B

26. What is an index in a database?
A. A structure that helps retrieve records faster
B. A type of database user
C. A backup file
D. A programming language
Answer: A
27. Which SQL command creates an index?
A. CREATE INDEX
B. ADD INDEX TABLE
C. INSERT INDEX
D. MAKE INDEX
Answer: A
28. Which command removes an index in MySQL?
A. DELETE INDEX
B. DROP INDEX
C. REMOVE INDEX
D. CLEAR INDEX
Answer: B
29. What is the main disadvantage of creating too many indexes?
A. Slower data modification operations
B. Automatic data deletion
C. Database shutdown
D. Loss of primary keys
Answer: A
30. Which type of index does a PRIMARY KEY create in an InnoDB table?
A. FULLTEXT
B. SPATIAL
C. Clustered index
D. Hash index
Answer: C

Part 4: Stored Procedures, Functions and Triggers

31. What is a stored procedure?
A. A saved collection of SQL statements
B. A database table
C. A hardware component
D. An operating system
Answer: A
32. Which SQL statement creates a stored procedure in MySQL?
A. CREATE PROCEDURE
B. MAKE PROCEDURE
C. INSERT PROCEDURE
D. ADD PROCEDURE
Answer: A
33. Which command executes a stored procedure in MySQL?
A. RUN
B. EXECUTE
C. CALL
D. START
Answer: C
34. Which SQL statement deletes a stored procedure?
A. DELETE PROCEDURE
B. DROP PROCEDURE

C. REMOVE PROCEDURE
D. CLEAR PROCEDURE
Answer: B
35. What is a database function?
A. A stored program that returns a value
B. A database table
C. A database user
D. A backup file
Answer: A
36. Which SQL statement creates a function in MySQL?
A. CREATE FUNCTION
B. MAKE FUNCTION
C. INSERT FUNCTION
D. ADD FUNCTION
Answer: A
37. Which keyword is used to return a value from a MySQL stored function?
A. OUTPUT
B. RETURN
C. SEND
D. RESULT
Answer: B
38. What is a database trigger?
A. A program that executes automatically when a specified database event occurs
B. A database password
C. A table column
D. A backup system
Answer: A
39. Which events can activate a MySQL trigger?
A. INSERT
B. UPDATE
C. DELETE
D. All of the above
Answer: D
40. Which SQL statement creates a trigger?
A. CREATE TRIGGER
B. MAKE TRIGGER
C. INSERT TRIGGER
D. ADD TRIGGER
Answer: A

Part 5: Database Object Management and Practical SQL

41. Which keyword refers to the new row inside a MySQL trigger?
A. OLD
B. NEW
C. CURRENT
D. NEXT
Answer: B
42. Which keyword refers to the previous row inside a MySQL UPDATE or DELETE trigger?
A. NEW
B. OLD
C. BEFORE
D. AFTER
Answer: B

43. How many times does a row-level trigger execute for each affected row?
A. Once for the entire table
B. Once for each affected row
C. Twice for each row
D. Only during SELECT
Answer: B
44. Which command displays stored procedures in MySQL?
A. SHOW PROCEDURE STATUS;
B. SHOW TABLES;
C. SHOW DATABASES;
D. SHOW USERS;
Answer: A
45. Which command displays triggers in MySQL?
A. SHOW TRIGGERS;
B. SHOW TABLES;
C. SHOW FUNCTIONS;
D. SHOW INDEXES;
Answer: A
46. Which command displays indexes of a table?
A. SHOW INDEX FROM table_name;
B. SHOW TABLE FROM index;
C. DISPLAY INDEX;
D. LIST INDEX;
Answer: A
47. Which SQL command changes the name of a column in MySQL?
A. ALTER TABLE ... RENAME COLUMN
B. UPDATE COLUMN
C. CHANGE DATABASE
D. INSERT COLUMN
Answer: A
48. Which command is used to remove all records from a table while keeping its structure?
A. DROP TABLE
B. TRUNCATE TABLE
C. DROP DATABASE
D. ALTER DATABASE
Answer: B
49. Which MySQL storage engine supports transactions and foreign keys?
A. MyISAM
B. InnoDB
C. CSV
D. MEMORY
Answer: B
50. Which of the following is an important task in database object management?
A. Creating tables
B. Managing views and indexes
C. Managing procedures and triggers
D. All of the above
Answer: D

MCQ Questions and Answers on Perform Data Manipulation Using SQL

Part 1: SQL and Data Manipulation Fundamentals

1. What does DML stand for?
A. Data Management Language
B. Data Manipulation Language
C. Database Management Logic
D. Data Modeling Language
Answer: B

2. Which of the following is a DML command?
A. CREATE
B. ALTER
C. INSERT
D. DROP
Answer: C

3. What is the main purpose of DML?
A. To manipulate data in database tables
B. To create an operating system
C. To configure network devices
D. To design web pages
Answer: A

4. Which SQL command retrieves data from a table?
A. INSERT
B. SELECT
C. UPDATE
D. DELETE
Answer: B

5. Which SQL command inserts new records into a table?
A. ADD
B. CREATE
C. INSERT
D. ALTER
Answer: C

6. Which SQL command modifies existing records?
A. UPDATE
B. INSERT
C. DROP
D. CREATE
Answer: A

7. Which SQL command removes records from a table?
A. REMOVE
B. DROP
C. DELETE
D. ALTER
Answer: C

8. Which clause specifies the table from which data is retrieved?
A. WHERE
B. FROM
C. ORDER BY
D. GROUP BY
Answer: B

9. Which symbol is used to select all columns from a table?

A. #
B. @
C. *
D. &
Answer: C
10. Which SQL clause filters records based on a condition?
A. ORDER BY
B. WHERE
C. GROUP BY
D. DISTINCT
Answer: B
Part 2: INSERT, UPDATE and DELETE
11. Which is the correct syntax for inserting a record?
A. ADD INTO student VALUES (1, 'Rahim');
B. INSERT INTO student VALUES (1, 'Rahim');
C. CREATE student VALUES (1, 'Rahim');
D. UPDATE student VALUES (1, 'Rahim');
Answer: B
12. Which SQL command inserts multiple records into a table?
A. INSERT INTO
B. UPDATE
C. DELETE
D. ALTER
Answer: A
13. Which keyword is used to modify existing records?
A. CHANGE
B. MODIFY
C. UPDATE
D. ALTER
Answer: C
14. What happens if UPDATE is executed without a WHERE clause?
A. Only one row is updated
B. All rows may be updated
C. No row is updated
D. The table is deleted
Answer: B
15. Which SQL statement deletes a specific record?
A. DELETE FROM student WHERE id = 5;
B. DROP student WHERE id = 5;
C. REMOVE student WHERE id = 5;
D. ALTER student WHERE id = 5;
Answer: A
16. What happens if DELETE is executed without a WHERE clause?
A. Only the first row is deleted
B. All rows are deleted
C. The table structure is deleted
D. Nothing happens
Answer: B
17. Which command removes all rows but keeps the table structure?
A. DROP
B. DELETE DATABASE

C. TRUNCATE TABLE
D. ALTER TABLE
Answer: C
18. Which SQL command is used to insert data from one table into another?
A. INSERT INTO ... SELECT
B. UPDATE ... SELECT
C. DELETE ... SELECT
D. DROP ... SELECT
Answer: A
19. Which keyword prevents duplicate rows in a SELECT result?
A. UNIQUE
B. DISTINCT
C. DIFFERENT
D. GROUP
Answer: B
20. Which operator is used to check whether a value falls within a range?
A. LIKE
B. IN
C. BETWEEN
D. IS NULL
Answer: C
Part 3: Filtering, Sorting and Searching
21. Which clause sorts query results?
A. GROUP BY
B. ORDER BY
C. WHERE
D. HAVING
Answer: B
22. Which keyword sorts data in ascending order?
A. DESC
B. ASC
C. SORT
D. UP
Answer: B
23. Which keyword sorts data in descending order?
A. ASC
B. DESC
C. DOWN
D. ORDER
Answer: B
24. Which operator is used for pattern matching in SQL?
A. BETWEEN
B. LIKE
C. IN
D. AND
Answer: B
25. Which wildcard represents any number of characters in SQL LIKE?
A. _
B. %
C. *
D. #
Answer: B

26. Which wildcard represents exactly one character in SQL LIKE?
A. %
B. *
C. _
D. #
Answer: C
27. Which operator checks whether a value matches any value in a specified list?
A. BETWEEN
B. IN
C. LIKE
D. EXISTS
Answer: B
28. Which operator combines two conditions where both must be true?
A. OR
B. NOT
C. AND
D. IN
Answer: C
29. Which operator returns records when at least one condition is true?
A. AND
B. OR
C. NOT
D. BETWEEN
Answer: B
30. Which operator is used to check NULL values?
A. = NULL
B. IS NULL
C. == NULL
D. LIKE NULL
Answer: B

Part 4: Aggregate Functions and Grouping

31. Which SQL function counts the number of rows?
A. SUM()
B. AVG()
C. COUNT()
D. MAX()
Answer: C
32. Which function calculates the average value?
A. AVG()
B. SUM()
C. COUNT()
D. MIN()
Answer: A
33. Which function calculates the total of numeric values?
A. MAX()
B. SUM()
C. AVG()
D. COUNT()
Answer: B
34. Which function returns the highest value?
A. MIN()
B. AVG()

C. MAX()
D. SUM()
Answer: C
35. Which function returns the lowest value?
A. MIN()
B. MAX()
C. SUM()
D. COUNT()
Answer: A
36. Which clause groups rows having the same values?
A. ORDER BY
B. GROUP BY
C. WHERE
D. DISTINCT
Answer: B
37. Which clause filters grouped records?
A. WHERE
B. ORDER BY
C. HAVING
D. FROM
Answer: C
38. Which SQL clause is generally used before GROUP BY?
A. WHERE
B. HAVING
C. ORDER BY
D. LIMIT
Answer: A
39. Which function counts only non-NULL values in a specified column?
A. COUNT(column_name)
B. SUM(column_name)
C. AVG(column_name)
D. MAX(column_name)
Answer: A
40. Which function returns the current date in MySQL?
A. NOW()
B. CURDATE()
C. TIME()
D. DATEONLY()
Answer: B
Part 5: Joins and Transactions
41. Which JOIN returns only matching records from both tables?
A. LEFT JOIN
B. RIGHT JOIN
C. INNER JOIN
D. CROSS JOIN
Answer: C
42. Which JOIN returns all records from the left table and matching records from the right table?
A. INNER JOIN
B. LEFT JOIN
C. CROSS JOIN
D. SELF JOIN
Answer: B

43. Which JOIN returns all possible combinations of rows from two tables?
A. INNER JOIN
B. LEFT JOIN
C. CROSS JOIN
D. RIGHT JOIN
Answer: C
44. Which JOIN is used to join a table with itself?
A. SELF JOIN
B. CROSS JOIN
C. LEFT JOIN
D. FULL JOIN
Answer: A
45. Which SQL statement is used to combine records from two tables based on a related column?
A. JOIN
B. DROP
C. ALTER
D. TRUNCATE
Answer: A
46. What does TCL stand for?
A. Table Control Language
B. Transaction Control Language
C. Technical Command Language
D. Transaction Creation Logic
Answer: B
47. Which command permanently saves a transaction?
A. ROLLBACK
B. SAVEPOINT
C. COMMIT
D. DELETE
Answer: C
48. Which command reverses uncommitted transaction changes?
A. COMMIT
B. ROLLBACK
C. SELECT
D. GRANT
Answer: B
49. Which command creates a point within a transaction to which you can roll back?
A. SAVEPOINT
B. COMMIT
C. DROP
D. UPDATE
Answer: A
50. What does ACID stand for in database transactions?
A. Atomicity, Consistency, Isolation, Durability
B. Access, Control, Integrity, Data
C. Automatic, Consistent, Internal, Database
D. Atomic, Computer, Internet, Data
Answer: A

MCQ Questions and Answers on Apply Advanced SQL Techniques

Part 1: Advanced SQL Fundamentals

1. What is Advanced SQL?
A. Basic data entry
B. Advanced techniques for querying and managing databases
C. Operating system management
D. Web page designing
Answer: B

2. Which SQL feature is used to write a query inside another query?
A. JOIN
B. Subquery
C. INDEX
D. TRIGGER
Answer: B

3. What is a subquery?
A. A query inside another SQL query
B. A database table
C. A database user
D. A backup file
Answer: A

4. Which operator is commonly used with a subquery that returns multiple values?
A. =
B. IN
C. +
D. -
Answer: B

5. Which operator checks whether a subquery returns at least one row?
A. EXISTS
B. BETWEEN
C. LIKE
D. DISTINCT
Answer: A

6. What is a correlated subquery?
A. A subquery that depends on the outer query
B. A query without a table
C. A query that deletes records
D. A query that creates a database
Answer: A

7. Which keyword is used to combine multiple query results and remove duplicates?
A. UNION
B. JOIN
C. GROUP BY
D. ORDER BY
Answer: A

8. Which operator combines query results and retains duplicates?
A. UNION
B. UNION ALL
C. DISTINCT
D. INTERSECT
Answer: B

9. Which keyword is used to remove duplicate rows from query results?

A. UNIQUE
B. DISTINCT
C. DIFFERENT
D. REMOVE
Answer: B
10. Which clause is used to limit the number of returned rows in MySQL?
A. TOP
B. LIMIT
C. RANGE
D. COUNT
Answer: B

Part 2: Advanced JOIN Operations

11. What is an INNER JOIN?
A. Returns only matching rows from both tables
B. Returns all rows from the first table
C. Returns all rows from both tables
D. Deletes duplicate records
Answer: A
12. Which JOIN returns all rows from the left table and matching rows from the right table?
A. INNER JOIN
B. LEFT JOIN
C. CROSS JOIN
D. SELF JOIN
Answer: B
13. What is a SELF JOIN?
A. Joining a table with itself
B. Joining three databases
C. Joining two servers
D. Joining a database with a file
Answer: A
14. Which JOIN produces a Cartesian product?
A. INNER JOIN
B. LEFT JOIN
C. CROSS JOIN
D. SELF JOIN
Answer: C
15. Which JOIN can be used to find records that do not have matching records in another table?
A. INNER JOIN
B. LEFT JOIN with IS NULL
C. CROSS JOIN
D. SELF JOIN
Answer: B
16. Which clause specifies the condition for joining tables?
A. WHERE
B. ON
C. GROUP BY
D. HAVING
Answer: B
17. What is an alias in SQL?
A. A temporary name for a table or column
B. A permanent database name

C. A password
D. A database backup
Answer: A
18. Which keyword assigns an alias to a column?
A. AS
B. ON
C. IN
D. BY
Answer: A

Part 3: Views, CTEs and Window Functions

19. What is a Common Table Expression (CTE)?
A. A temporary named result set used within a single SQL statement
B. A permanent database
C. A backup table
D. A database password
Answer: A
20. Which keyword defines a CTE in MySQL?
A. CREATE
B. WITH
C. SELECT
D. TEMP
Answer: B
21. What is a recursive CTE used for?
A. Processing hierarchical or recursive data
B. Deleting databases
C. Creating users
D. Encrypting tables
Answer: A
22. What is a window function?
A. A function that performs calculations across related rows without collapsing them into one row
B. A function that creates a database
C. A function that deletes tables
D. A function that backs up data
Answer: A
23. Which clause is used with window functions to define a window?
A. GROUP BY
B. OVER()
C. WHERE
D. HAVING
Answer: B
24. Which function assigns a unique sequential number to rows?
A. RANK()
B. DENSE_RANK()
C. ROW_NUMBER()
D. COUNT()
Answer: C
25. Which function assigns ranks with gaps when there are ties?
A. ROW_NUMBER()
B. RANK()
C. DENSE_RANK()
D. SUM()
Answer: B

26. Which function assigns ranks without gaps when there are ties?
A. DENSE_RANK()
B. ROW_NUMBER()
C. COUNT()
D. AVG()
Answer: A
27. Which window function calculates a running total?
A. SUM() OVER()
B. COUNT()
C. MAX()
D. MIN()
Answer: A
28. Which window function accesses the value from the previous row?
A. LEAD()
B. LAG()
C. FIRST_VALUE()
D. LAST_VALUE()
Answer: B
29. Which window function accesses the value from the next row?
A. LAG()
B. LEAD()
C. RANK()
D. SUM()
Answer: B
30. Which clause divides rows into groups for window calculations?
A. PARTITION BY
B. GROUP BY
C. ORDER BY
D. WHERE
Answer: A

Part 4: Stored Procedures, Functions and Triggers

31. What is a stored procedure?
A. A collection of SQL statements stored in the database
B. A database table
C. A data type
D. A database password
Answer: A
32. Which statement is used to execute a stored procedure in MySQL?
A. RUN
B. CALL
C. START
D. OPEN
Answer: B
33. Which keyword is used to return a value from a stored function?
A. OUTPUT
B. RETURN
C. RESULT
D. SEND
Answer: B
34. Which parameter mode is used to pass a value into a stored procedure?
A. OUT
B. IN

C. BOTH
D. RETURN
Answer: B
35. Which parameter mode returns a value from a stored procedure?
A. IN
B. OUT
C. SELECT
D. WHERE
Answer: B
36. What is a trigger?
A. A stored program that executes automatically when a specified table event occurs
B. A database user
C. A database column
D. A backup file
Answer: A
37. Which events can activate a MySQL trigger?
A. INSERT
B. UPDATE
C. DELETE
D. All of the above
Answer: D
38. Which keyword refers to the new row in a MySQL INSERT trigger?
A. OLD
B. NEW
C. NEXT
D. CURRENT
Answer: B
39. Which keyword refers to the old row in a MySQL UPDATE trigger?
A. NEW
B. OLD
C. BEFORE
D. AFTER
Answer: B
40. Which SQL command removes a stored procedure?
A. DELETE PROCEDURE
B. DROP PROCEDURE
C. REMOVE PROCEDURE
D. CLEAR PROCEDURE
Answer: B

Part 5: Indexes, Transactions and Query Optimization

41. What is the main purpose of an index?
A. To improve data retrieval performance
B. To delete records
C. To increase duplicate values
D. To create users
Answer: A
42. Which SQL statement creates a composite index?
A. CREATE INDEX idx_name ON student(name, age);
B. CREATE TABLE idx_name;
C. INSERT INDEX student;
D. ALTER DATABASE idx_name;
Answer: A

43. What is a transaction in SQL?
A. A group of database operations treated as a single logical unit
B. A database table
C. A programming language
D. A backup file
Answer: A
44. Which command saves transaction changes permanently?
A. ROLLBACK
B. COMMIT
C. SAVEPOINT
D. DELETE
Answer: B
45. Which command cancels uncommitted transaction changes?
A. COMMIT
B. UPDATE
C. ROLLBACK
D. INSERT
Answer: C
46. What does ACID stand for?
A. Atomicity, Consistency, Isolation, Durability
B. Access, Control, Integrity, Data
C. Automatic, Consistent, Internal, Database
D. Atomic, Computer, Internet, Database
Answer: A
47. Which SQL command creates a savepoint within a transaction?
A. CREATE SAVEPOINT
B. SAVEPOINT
C. COMMIT POINT
D. SET COMMIT
Answer: B
48. What is query optimization?
A. Improving SQL query execution efficiency
B. Deleting database records
C. Increasing duplicate records
D. Removing database users
Answer: A
49. Which MySQL command helps analyze how a query will be executed?
A. DESCRIBE only
B. EXPLAIN
C. DELETE
D. UPDATE
Answer: B
50. Which technique can improve the performance of a frequently filtered column?
A. Creating an appropriate index
B. Adding duplicate records
C. Removing all constraints
D. Using SELECT * in every query
Answer: A

MCQ Questions and Answers on Configure Database Security

Part 1: Database Security Fundamentals

1. What is database security?
A. Managing computer hardware
B. Protecting database information from unauthorized access
C. Creating websites
D. Installing operating systems
Answer: B

2. What is the main purpose of database security?
A. To increase duplicate data
B. To protect data confidentiality, integrity, and availability
C. To delete unnecessary records
D. To increase database size
Answer: B

3. What does CIA stand for in information security?
A. Confidentiality, Integrity, Availability
B. Control, Information, Access
C. Computer, Internet, Application
D. Confidentiality, Internet, Authentication
Answer: A

4. Which security principle ensures that only authorized users can access data?
A. Availability
B. Confidentiality
C. Redundancy
D. Scalability
Answer: B

5. Which security principle ensures that data remains accurate and unchanged without authorization?
A. Integrity
B. Availability
C. Authentication
D. Backup
Answer: A

6. What does authentication mean?
A. Verifying the identity of a user
B. Deleting database records
C. Creating tables
D. Encrypting files
Answer: A

7. What does authorization mean?
A. Verifying user identity
B. Determining what an authenticated user is allowed to do
C. Creating a database
D. Installing software
Answer: B

8. Which of the following is a common database security threat?
A. SQL Injection
B. Data backup
C. Database indexing
D. Normalization
Answer: A

9. What is the principle of least privilege?

A. Giving users all permissions
B. Giving users only the permissions necessary for their work
C. Allowing anonymous access
D. Sharing administrator passwords
Answer: B
10. Which of the following helps protect database access?
A. Strong passwords
B. User authentication
C. Access control
D. All of the above
Answer: D

Part 2: User and Privilege Management

11. Which SQL command creates a new MySQL user?
A. ADD USER
B. CREATE USER
C. INSERT USER
D. NEW USER
Answer: B
12. Which SQL command grants privileges to a user?
A. REVOKE
B. GRANT
C. DELETE
D. DROP
Answer: B
13. Which SQL command removes privileges from a user?
A. GRANT
B. CREATE
C. REVOKE
D. INSERT
Answer: C
14. Which SQL command deletes a MySQL user account?
A. DELETE USER
B. DROP USER
C. REMOVE USER
D. CLEAR USER
Answer: B
15. Which SQL command changes a user's password in MySQL?
A. ALTER USER
B. DELETE USER
C. DROP USER
D. SHOW USER
Answer: A
16. Which command displays the current MySQL user?
A. SELECT USER();
B. SHOW USERS;
C. DISPLAY USER;
D. GET USER;
Answer: A
17. Which privilege allows a user to retrieve data from a table?
A. INSERT
B. SELECT

C. DELETE
D. UPDATE
Answer: B
18. Which privilege allows a user to add new records?
A. SELECT
B. UPDATE
C. INSERT
D. DROP
Answer: C
19. Which privilege allows a user to modify existing records?
A. UPDATE
B. SELECT
C. CREATE
D. DROP
Answer: A
20. Which privilege allows a user to remove records?
A. INSERT
B. DELETE
C. SELECT
D. ALTER
Answer: B

Part 3: Password and Access Control Security

21. Which is the strongest password?
A. 123456
B. password
C. Admin123
D. T#9mQ!7zP@2x
Answer: D
22. What is the purpose of multi-factor authentication (MFA)?
A. To provide additional identity verification
B. To delete duplicate data
C. To increase database storage
D. To create tables
Answer: A
23. What is role-based access control (RBAC)?
A. Assigning permissions through user roles
B. Giving every user administrator access
C. Deleting database users
D. Sharing passwords
Answer: A
24. Which MySQL command creates a role?
A. CREATE ROLE
B. ADD ROLE
C. INSERT ROLE
D. NEW ROLE
Answer: A
25. Which command assigns a role to a user in MySQL?
A. GRANT role_name TO user_name;
B. INSERT ROLE user_name;
C. CREATE USER role_name;
D. DELETE ROLE user_name;
Answer: A

26. Which SQL command displays the privileges granted to a user?
A. SHOW PRIVILEGES
B. SHOW GRANTS
C. DISPLAY ACCESS
D. LIST PERMISSIONS
Answer: B
27. What is an access control list (ACL)?
A. A list of permissions controlling access to resources
B. A database backup file
C. A table structure
D. A programming language
Answer: A
28. What is the purpose of account locking?
A. To prevent an account from logging in
B. To delete the database
C. To increase storage
D. To create a new table
Answer: A
29. Which MySQL statement locks a user account?
A. ALTER USER 'user1'@'localhost' ACCOUNT LOCK;
B. DELETE USER 'user1';
C. DROP ACCOUNT user1;
D. LOCK DATABASE user1;
Answer: A
30. What is the main purpose of database auditing?
A. To monitor and record database activities
B. To delete user accounts
C. To create indexes
D. To increase database size
Answer: A

Part 4: Database Threats and Protection

31. What is SQL Injection?
A. An attack that inserts malicious SQL through application inputs
B. A database backup method
C. A data compression technique
D. A database normalization method
Answer: A
32. Which technique helps prevent SQL Injection?
A. Prepared statements
B. Weak passwords
C. Sharing database credentials
D. Disabling access control
Answer: A
33. What is encryption?
A. Converting readable data into an encoded form
B. Deleting database records
C. Creating database tables
D. Sorting database records
Answer: A
34. Which encryption method uses the same key for encryption and decryption?
A. Asymmetric encryption
B. Symmetric encryption

C. Hashing
D. Digital signature
Answer: B
35. Which protocol helps secure database communication over a network?
A. FTP
B. Telnet
C. TLS
D. HTTP
Answer: C
36. What is data masking?
A. Hiding sensitive data by replacing or obscuring values
B. Deleting all database records
C. Increasing database size
D. Creating a new database
Answer: A
37. What is a database firewall used for?
A. Monitoring and controlling database traffic
B. Formatting hard disks
C. Creating tables
D. Increasing memory
Answer: A
38. What is a brute-force attack?
A. Trying many possible passwords to gain access
B. Creating database indexes
C. Backing up database records
D. Normalizing tables
Answer: A
39. Which technique helps protect passwords stored in a database?
A. Storing passwords in plain text
B. Secure password hashing with salt
C. Sharing passwords
D. Using the same password for all users
Answer: B
40. Which security measure helps protect a database server from unauthorized network access?
A. Firewall rules
B. Weak passwords
C. Public access
D. Anonymous login
Answer: A

Part 5: Backup, Recovery and Security Maintenance

41. Why is database backup important for security?
A. To recover data after loss or damage
B. To create duplicate users
C. To delete database tables
D. To increase query complexity
Answer: A
42. Which MySQL utility is commonly used for logical database backup?
A. mysqldump
B. ping
C. ipconfig
D. netstat
Answer: A

43. Which command can restore a MySQL SQL backup?
A. mysql -u root -p database_name < backup.sql
B. DROP DATABASE database_name;
C. DELETE DATABASE database_name;
D. SHOW TABLES;
Answer: A
44. What is disaster recovery?
A. Restoring systems and data after a major failure
B. Creating new user accounts
C. Deleting database records
D. Changing table names
Answer: A
45. What is the purpose of database security updates?
A. To fix vulnerabilities and improve security
B. To increase duplicate records
C. To remove all users
D. To delete backups
Answer: A
46. What is data confidentiality?
A. Ensuring data is accessible only to authorized parties
B. Allowing everyone to access data
C. Deleting sensitive information
D. Increasing data duplication
Answer: A
47. Which database security practice is recommended?
A. Regularly reviewing user privileges
B. Sharing administrator passwords
C. Disabling backups
D. Allowing anonymous access
Answer: A
48. What is an unauthorized access attempt?
A. Accessing a database without proper permission
B. Creating a database backup
C. Running an authorized query
D. Updating a permitted record
Answer: A
49. Which MySQL command displays the current authenticated account?
A. SELECT CURRENT_USER();
B. SHOW DATABASES;
C. SELECT VERSION();
D. SHOW TABLES;
Answer: A
50. Which of the following is the best overall database security strategy?
A. Strong authentication
B. Access control and least privilege
C. Regular backups and security updates
D. All of the above
Answer: D

MCQ Questions and Answers on Perform Database Maintenance

Section A: Database Maintenance Fundamentals

1. What is database maintenance?
A. Creating a website
B. Keeping a database efficient, secure, and reliable
C. Designing graphics
D. Installing an operating system
Answer: B

2. What is the main purpose of database maintenance?
A. To increase database errors
B. To improve database performance and reliability
C. To delete all records
D. To stop database services
Answer: B

3. Which command is used to check a MySQL table?
A. CHECK TABLE
B. VERIFY TABLE
C. TEST TABLE
D. VALIDATE TABLE
Answer: A

4. Which command is used to optimize a MySQL table?
A. REPAIR TABLE
B. OPTIMIZE TABLE
C. UPDATE TABLE
D. MODIFY TABLE
Answer: B

5. Which command is used to repair a MySQL table?
A. FIX TABLE
B. REPAIR TABLE
C. RESTORE TABLE
D. RESET TABLE
Answer: B

6. Which storage engine supports the REPAIR TABLE operation?
A. InnoDB
B. MyISAM
C. Memory
D. CSV
Answer: B

7. What is database fragmentation?
A. Organizing database users
B. Inefficient storage of data
C. Creating a new database
D. Deleting database users
Answer: B

8. Which activity helps improve database performance?
A. Removing all indexes
B. Query optimization
C. Increasing duplicate data
D. Disabling backups
Answer: B

9. What is database monitoring?

A. Tracking database performance and activities
B. Deleting database tables
C. Formatting a hard disk
D. Creating web pages
Answer: A
10. Which MySQL command displays database information?
A. SHOW DATABASES;
B. DISPLAY DATABASE;
C. LIST DATABASE;
D. VIEW DATABASE;
Answer: A

Section B: Database Backup and Recovery

11. What is a database backup?
A. Deleting database records
B. Creating a copy of database data
C. Changing a database name
D. Removing database users
Answer: B
12. Which MySQL utility is used to create a logical backup?
A. mysql
B. mysqldump
C. mysqladmin
D. mysqlcheck
Answer: B
13. Which command restores a MySQL database from an SQL backup file?
A. mysql -u root -p database_name < backup.sql
B. mysqldump database_name
C. DROP DATABASE database_name
D. SHOW DATABASES
Answer: A
14. What is a full backup?
A. Backup of selected records
B. Backup of the entire database
C. Backup of only indexes
D. Backup of only users
Answer: B
15. What is an incremental backup?
A. Backup of all data every time
B. Backup of changes since the previous backup
C. Backup of database structure only
D. Backup of deleted records only
Answer: B
16. What is a differential backup?
A. Backup of changes since the last full backup
B. Backup of only database users
C. Backup of all server applications
D. Backup of only indexes
Answer: A
17. Why should database backups be tested?
A. To increase database size
B. To ensure data can be restored successfully

C. To remove duplicate records
D. To change table structure
Answer: B
18. What does database recovery mean?
A. Restoring data after failure or loss
B. Creating a new user
C. Deleting old tables
D. Changing a database password
Answer: A
19. Which MySQL feature helps recover data using recorded changes?
A. Binary Log
B. Primary Key
C. Foreign Key
D. View
Answer: A
20. What is the purpose of a transaction log?
A. To record database changes and transactions
B. To store images
C. To create user accounts
D. To format database tables
Answer: A
Section C: Database Performance Optimization
21. What is an index in a database?
A. A tool for faster data retrieval
B. A database backup
C. A database password
D. A type of database user
Answer: A
22. Which SQL command creates an index?
A. CREATE INDEX
B. ADD INDEX TABLE
C. MAKE INDEX
D. INSERT INDEX
Answer: A
23. Which SQL command removes an index in MySQL?
A. DELETE INDEX
B. DROP INDEX
C. REMOVE INDEX
D. CLEAR INDEX
Answer: B
24. What is the purpose of EXPLAIN in MySQL?
A. To analyze query execution plans
B. To delete a table
C. To create a database
D. To restore a backup
Answer: A
25. Which command displays the structure of a table?
A. DESCRIBE table_name;
B. DELETE table_name;
C. DROP table_name;
D. UPDATE table_name;
Answer: A

26. What is a slow query?
A. A query that takes a long time to execute
B. A query that creates a database
C. A query that deletes users
D. A query that has no conditions
Answer: A
27. Which MySQL feature helps identify slow queries?
A. Slow Query Log
B. Error Table
C. User Table
D. Primary Key
Answer: A
28. What is query optimization?
A. Improving SQL query execution efficiency
B. Increasing duplicate records
C. Deleting all indexes
D. Removing database constraints
Answer: A
29. Which command shows the currently running MySQL processes?
A. SHOW PROCESSLIST;
B. SHOW DATABASES;
C. SHOW TABLES;
D. SHOW INDEXES;
Answer: A
30. What is the main purpose of database caching?
A. To improve data access speed
B. To delete old data
C. To create new users
D. To increase data duplication
Answer: A

Section D: Database Security and Monitoring

31. Which command is used to change a MySQL user password?
A. ALTER USER
B. CHANGE PASSWORD TABLE
C. UPDATE PASSWORD
D. MODIFY DATABASE
Answer: A
32. Which command displays MySQL user accounts?
A. SELECT User, Host FROM mysql.user;
B. SHOW USER TABLE;
C. DISPLAY USERS;
D. LIST USER;
Answer: A
33. What is the purpose of database auditing?
A. Tracking database activities
B. Creating duplicate databases
C. Removing indexes
D. Formatting storage devices
Answer: A
34. What is a database log file?
A. A file that records database events or activities
B. A file containing only images

C. A file containing operating system drivers
D. A file used for designing websites
Answer: A
35. Which command displays the MySQL server status?
A. SHOW STATUS;
B. SHOW TABLE;
C. SHOW USER;
D. SHOW INDEX;
Answer: A
36. What is database availability?
A. The ability of a database to remain accessible
B. The number of database tables
C. The size of a database file
D. The number of SQL queries
Answer: A
37. What is database downtime?
A. The period when a database is unavailable
B. The period when a database is running normally
C. The time taken to create a table
D. The time taken to insert a record
Answer: A
38. Why should database software be updated?
A. To improve security and fix bugs
B. To delete all records
C. To disable database access
D. To increase errors
Answer: A
39. Which practice helps protect a database from unauthorized access?
A. Strong passwords and access control
B. Sharing passwords
C. Disabling authentication
D. Giving everyone administrator privileges
Answer: A
40. What is database integrity?
A. Accuracy and consistency of data
B. Database size
C. Number of database users
D. Number of database servers
Answer: A
Section E: Practical Database Maintenance
41. Which command is used to display the size of a MySQL database?
A. SELECT with information_schema
B. DELETE DATABASE
C. DROP DATABASE
D. CREATE DATABASE
Answer: A
42. Which MySQL system database stores server-related information?
A. information_schema
B. student
C. employee
D. customer
Answer: A

43. Which command removes all records from a table while keeping its structure?
A. DROP TABLE
B. TRUNCATE TABLE
C. REMOVE DATABASE
D. DELETE DATABASE
Answer: B
44. Which command deletes selected records from a table?
A. DELETE
B. CREATE
C. ALTER
D. GRANT
Answer: A
45. What is the purpose of OPTIMIZE TABLE?
A. To reorganize table storage and reclaim space where supported
B. To delete the database
C. To create a new user
D. To change the database password
Answer: A
46. Which MySQL utility can check and optimize database tables?
A. mysqlcheck
B. notepad
C. ping
D. ipconfig
Answer: A
47. What should be done before performing major database maintenance?
A. Take a backup
B. Delete all indexes
C. Remove all users
D. Shut down the computer permanently
Answer: A
48. What is database archiving?
A. Moving older data to long-term storage
B. Deleting the database permanently
C. Changing the database password
D. Creating duplicate users
Answer: A
49. What is the purpose of database capacity planning?
A. Estimating future storage and resource requirements
B. Deleting database records
C. Creating SQL queries
D. Changing table names
Answer: A
50. Which activity is an important part of regular database maintenance?
A. Backup, monitoring, security updates, and performance checks
B. Deleting all database tables
C. Disabling database logs
D. Removing all constraints
Answer: A

MCQ Questions and Answers on Manage Database on Cloud

Section A: Cloud Database Fundamentals

1. What is a cloud database?
A. A database stored only on a local computer
B. A database hosted on cloud infrastructure
C. A database stored on a USB drive
D. A database without a server
Answer: B

2. What is cloud computing?
A. Delivering computing services over the internet
B. Installing software from a CD
C. Creating computer hardware
D. Formatting a hard disk
Answer: A

3. Which of the following is a cloud service provider?
A. Microsoft Azure
B. Notepad
C. VLC Player
D. MS Paint
Answer: A

4. Which of the following is a cloud database service?
A. Amazon RDS
B. MS Word
C. Adobe Photoshop
D. Windows Media Player
Answer: A

5. What does DBaaS stand for?
A. Database Backup as a Service
B. Database as a Service
C. Data Backup and Storage
D. Database Application Security
Answer: B

6. Which company provides Google Cloud SQL?
A. Microsoft
B. Amazon
C. Google
D. IBM
Answer: C

7. What is the main advantage of a cloud database?
A. Limited accessibility
B. Remote accessibility
C. No data storage
D. No internet requirement
Answer: B

8. Which cloud deployment model is dedicated to a single organization?
A. Public Cloud
B. Private Cloud
C. Community Cloud
D. Hybrid Cloud
Answer: B

9. What is a hybrid cloud?

A. Combination of public and private cloud
B. Combination of two local computers
C. A database without internet
D. A single physical server
Answer: A
10. Which cloud deployment model provides services to multiple customers over shared infrastructure?
A. Private Cloud
B. Public Cloud
C. Local Cloud
D. Personal Cloud
Answer: B

Section B: Cloud Database Providers and Services

11. What does AWS stand for?
A. Advanced Web System
B. Amazon Web Services
C. Application Web Server
D. Automated Web Software
Answer: B
12. Which AWS service provides managed relational databases?
A. Amazon RDS
B. Amazon S3
C. Amazon EC2
D. Amazon CloudFront
Answer: A
13. Which AWS database service is designed for NoSQL workloads?
A. Amazon RDS
B. Amazon DynamoDB
C. Amazon S3
D. Amazon Route 53
Answer: B
14. What is Microsoft Azure SQL Database?
A. A managed cloud relational database service
B. A text editor
C. An operating system
D. A web browser
Answer: A
15. Which Google Cloud service provides managed MySQL databases?
A. Google Cloud Storage
B. Cloud SQL
C. Google Drive
D. Google Docs
Answer: B
16. What is Amazon Aurora?
A. A relational database service
B. A cloud storage folder
C. A programming language
D. A web browser
Answer: A
17. Which service is used for cloud object storage in AWS?
A. Amazon RDS
B. Amazon S3

C. Amazon EC2
D. Amazon VPC
Answer: B
18. What is the purpose of Amazon EC2?
A. Providing virtual servers
B. Creating SQL queries
C. Managing email accounts
D. Designing databases
Answer: A
19. Which of the following is a managed cloud database platform?
A. Azure SQL Database
B. Microsoft Paint
C. Windows Notepad
D. Google Chrome
Answer: A
20. What does RDS stand for?
A. Relational Database Service
B. Remote Data Software
C. Rapid Database System
D. Relational Data Storage
Answer: A

Section C: Cloud Database Deployment and Configuration

21. What is database provisioning?
A. Creating and allocating database resources
B. Deleting database records
C. Formatting a computer
D. Removing database users
Answer: A
22. What is a cloud database instance?
A. A running database environment in the cloud
B. A database password
C. A SQL command
D. A database table
Answer: A
23. Which factor should be considered when selecting a cloud database?
A. Performance requirements
B. Security requirements
C. Cost and scalability
D. All of the above
Answer: D
24. What is scalability in cloud computing?
A. Ability to increase or decrease resources
B. Ability to delete databases
C. Ability to remove users
D. Ability to stop the internet
Answer: A
25. What is vertical scaling?
A. Adding more CPU or RAM to an existing server
B. Adding more users
C. Creating more database tables
D. Moving data to another country
Answer: A

26. What is horizontal scaling?
A. Increasing the capacity of one server only
B. Adding more servers or database instances
C. Deleting database tables
D. Reducing storage capacity
Answer: B
27. What is cloud database migration?
A. Moving a database to a cloud environment
B. Deleting database records
C. Changing a table name
D. Creating a local user
Answer: A
28. Which protocol is commonly used for secure database administration over a network?
A. FTP
B. SSH
C. HTTP
D. SMTP
Answer: B
29. Which port is the default MySQL port?
A. 80
B. 443
C. 3306
D. 8080
Answer: C
30. What is the purpose of a cloud firewall or security group?
A. Controlling network traffic
B. Creating database tables
C. Writing SQL queries
D. Increasing database records
Answer: A

Section D: Cloud Database Security and Backup

31. What is encryption?
A. Converting data into a protected format
B. Deleting database records
C. Creating database tables
D. Increasing database size
Answer: A
32. What does encryption at rest mean?
A. Protecting stored data
B. Protecting data during transmission only
C. Deleting stored data
D. Compressing database files
Answer: A
33. What does encryption in transit mean?
A. Protecting data while it is transmitted
B. Protecting data stored on a disk only
C. Deleting network traffic
D. Creating database backups
Answer: A
34. Which protocol is commonly used to secure data transmission?
A. HTTP
B. HTTPS/TLS

C. FTP
D. Telnet
Answer: B
35. What is multi-factor authentication (MFA)?
A. Using multiple verification methods
B. Using multiple databases
C. Creating multiple tables
D. Using multiple SQL commands
Answer: A
36. What is the principle of least privilege?
A. Giving users only the permissions they need
B. Giving all users administrator access
C. Sharing passwords
D. Disabling security settings
Answer: A
37. What is an automated backup?
A. A backup performed automatically according to a schedule
B. A backup performed only manually
C. A backup without data
D. A backup that deletes the database
Answer: A
38. What is a database snapshot?
A. A point-in-time copy of database storage or state
B. A database password
C. A database query
D. A database user account
Answer: A
39. What is point-in-time recovery (PITR)?
A. Restoring a database to a specific previous time
B. Creating a new database user
C. Deleting all database logs
D. Increasing database storage
Answer: A
40. Why should cloud database backups be stored securely?
A. To prevent unauthorized access and data loss
B. To increase database errors
C. To reduce database performance
D. To disable database recovery
Answer: A

Section E: Cloud Database Monitoring and Management

41. What is cloud database monitoring?
A. Tracking database health and performance
B. Deleting database tables
C. Creating a website
D. Installing an operating system
Answer: A
42. Which metric is commonly monitored in a cloud database?
A. CPU utilization
B. Memory usage
C. Storage usage
D. All of the above
Answer: D

43. What is high availability?
A. Keeping a database accessible with minimal downtime
B. Making a database unavailable
C. Deleting database backups
D. Disabling database services
Answer: A
44. What is database replication?
A. Copying data from one database server to another
B. Deleting database records
C. Changing a database password
D. Removing database indexes
Answer: A
45. What is a read replica?
A. A copy of a database used mainly to handle read queries
B. A database backup file only
C. A database administrator
D. A database password
Answer: A
46. What is load balancing?
A. Distributing workloads across multiple resources
B. Deleting database records
C. Creating database passwords
D. Disabling database connections
Answer: A
47. What is a Service Level Agreement (SLA)?
A. An agreement defining service expectations and commitments
B. A database table
C. A SQL query
D. A programming language
Answer: A
48. What is vendor lock-in?
A. Difficulty moving from one cloud provider to another
B. Locking a database table
C. Protecting a database password
D. Deleting cloud resources
Answer: A
49. What is cloud database cost optimization?
A. Managing resources to reduce unnecessary expenses
B. Increasing unused resources
C. Disabling all database backups
D. Buying unnecessary storage
Answer: A
50. Which activity is essential for managing a cloud database?
A. Monitoring, backup, security, and performance management
B. Deleting all database records
C. Disabling authentication
D. Removing all database users
Answer: A